The Collective Motion Using Computer Modelling

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1 Introduction

There are many collective motions in nature, from colonies of bacteria to swarms of birds flying in the sky. However, whatever the scale of the animals, they seem to follow the same fundamental principle behind these collective motions. One the reason why many animals or living organisms take collective motion is that it is more efficient to explore for food or other resources and another reason is that it is much easier to defend against predators in flocks.

In 1987, Craig Reynolds developed an algorithms to try to mimic the flocking of birds. He referes to flocks of birds, generally as ‘‘boid’’ or ‘‘bird-oid’’ objects [1]. He noticed that birds follow some simple rules that correspond to the motion of their immediate neighbours.

One of the similar models was developed in 1995, which is known as the Viscek model [2]. In this model, the self-driven particles will move in the same direction with small deviation from the nearest neighbouring particle. The group motion can undergo phase transitions, as was observed in the Viscek model. The factors governing individual motion, such as average velocity and strength of interactions, are varied.

In 1996, a mathematical model was proposed which categorized different types of collective motion patterns and obtain their phase diagram [3, 4].

The model in which the collective motion and formation of bacteria colony on a ‘‘solid agar surface’’ was studied specifically can be trace back to the 1996 [5]. The model simplified the bacteria as self-propelled particles and the interaction only change the direction of the movement. In addition, the paper investigate the formation of vortex observed in the colony. This is one of the earlier models that focus on describing the collective motion of bacteria.

Another model which investigate the swarming behaviour in biological system especially the Daphnia swarms [6]. The model of Brownian agent that was introduced in the paper can simulate real Daphina swarms and show reasonable swarming behaviour. The Daphnia will swim around a light source and form a voetex. The model demonstrate that vortex swarming does not required an explicit alignment rule, instead it can be archived by using attraction and repulsion [7]. In the model, it introduce avoidance behaviour. In the first approach of this simple avoidance model, it was assumed that the repulsive force between the two agents depends inversely on the distance between two agents [7]. Using the generalised Langevin equation, the modified equation of motion can be found by including the repulsive force. The second approach improved the model by reducing the sudden maneuvers of the agents. The formation of vortex observed in the simulation is the same as what we observed in the natural world in Daphnia.

Models which focus on self-propelled rods are categorized into two types of model on the basis of their interaction [8]. The first type of model is dry self-propelled rods, which is introduced for rod-shaped objects which drive through dissipative medium with repulsive interaction [9, 8].

In the paper that introduced self-propelled rod, it also proposed a model which demonstrate the relevance of the volume exclusion and self-propulsion for an effective alignment of the particles [4, 9]. The system described as ''dry'' is defined as those where momentum is not conserved due to friction with a substrate or an embedding porous medium [10]. The second type of model is wet self-propelled rods, which characterise short-ranged repulsion and long-ranged hydrodynamic interactions which is the result of a flow fields of particles [8].

There are many observations and experiments made for investigating the collective motion of bacteria. One of most recent experiment conducted by Ghosh and Cheng discovered that the collective dynamics have strong influence over individual behaviour of bacteria [11]. In addition, they found that when bacteria can cross over each other during an encounters a long-range nematic order emerges. When bacteria are prevented to cross over each other by a confinement, a short-range polar order was formed.

One of the recent advances was made in the investigation of behavioral mechanisms in swarming locusts, which discovered that locusts do not explicitly align with neighbours and classical models of collective models of collective behaviour cannot account for the formation of swarming in marching desert locusts [12]. In addition, there is no observation on density-dependent phase transition from disorder to ordered collective motion. Saying, et al. discovered that by using minimal cognitive model which is also known as ring attractor network it can explain the individual and collective bahaviour of locusts [12].

As mentioned above, there are many different types of models describe the collective motion behaviour observed in nature. However, since the same patterns was observed in the collective motion, there must be still to be discovered theories or laws which can describe all the collection motion observed in nature [4].

The Viscek model requires the particle to know its environment, whereas the bacterium that we are considering in this case does not know its environment. Therefore, we will not be using this method in our model.

In this report, we first use Langevin Leap-frog method of integration to simulate the path in which bacteria traveled. Next we consider the fundamental interaction between bacteria and its surrounding fluid into our model. A comparison between the result obtained from our model and the experiment observation is presented in the Discussion section.

2 Collective Motion Model

For only one bacterium, there are the self-drive force, random force, spring force and bending force of the spring acting on the bacterium. Once we calculated the force acting on those three spheres or the bacterium, we can use the Langevin Leap-frog method of integration to calculate the displacement for each time step [13]. Assume $v\left(t-\left(\frac{1}{2}\right)h\right)$, $x(t)$ and $F(t) = ma$ are the known velocity, displacement and force components, respectively, and a is the acceleration at time t. The impulsive Langevin extension of the leap-frog algorithm then can be written as:

$$v = v\left(t - \frac{1}{2}h\right) + ah \tag{2.1}$$

$$\Delta v = -fv + \sqrt{f(2-f)\left(\frac{k_B T_{\text{ref}}}{m}\right)}\xi \tag{2.2}$$

$$x(t+h) = x(t) + \left(v + \frac{1}{2}\Delta v\right)h \tag{2.3}$$

$$v\left(t + \frac{1}{2}h\right) = v + \Delta t \tag{2.4}$$

where h is the time step Δt. Equation 2.1 update the velocity in the leap-frog scheme. Equation 2.2 includes the impulsive application of the friction which can reduce the velocity by a fraction $f : 0 \leq f \leq 1$, and noise, ξ, is a random sample from a normal distribution. Equation 2.3 updates the coordinates, taking into account that Δv is applied only between $t + \left(\frac{1}{2}\right)h$ and $t+h$. Equation 2.3 can be considered in two additional steps, as it is shown in Equation 2.5 and Equation 2.6. Equation 2.4 gives the modified velocity to the end of the time step [13]. The leap-frog method is also known as the explicit central difference method.

$$x\left(t + \frac{1}{2}h\right) = x(t) + v\frac{1}{2}h \tag{2.5}$$

$$x(x+\Delta x) = x\left(t + \frac{1}{2}h\right) + (v+\Delta v)\frac{1}{2}h \tag{2.6}$$

To find the value of the fraction f, we assume that the total force is equal to zero and use the formula:

$$\lambda = -\frac{1}{h}\ln(1-f) \tag{2.7}$$

where λ is drag coefficient. The drag coefficient can be calculated using Stokes' law, which can be found as $\lambda = 6\pi R\eta$, where η is the viscosity and R is radius of the beads, which is equal to 0.5 µm in our model. This will give us the value of the fraction f.

2.1 Many Bacteria Model

We focus on the collective motion of bacteria, where we can use the Van der Waals force to represent the attractive force each bacterium experiences within swarms of bacteria. For two spheres with identical radius R, and the distance between the surfaces of these two spheres is l, where $l \ll R$,

$$U = -\frac{A_{\mathrm{Ham}}}{6}\frac{R}{2l} \tag{2.8}$$

where A_{Ham} is the Hamaker constant. This is the expression for the interation energy [14], U. The Van der Waals force can be found by taking the negative derivatives of Equation 2.8, which is $\boldsymbol{F} = -\boldsymbol{\nabla} U$. Since bacteria are mainly made of protein, the Hamaker constant that we will be using is 12.6×10^{-21}J, which is the value for the interaction between protein [15].

In the model where we involves many bacteria, we need to consider the hydrodynamic force. When the bacterium moving in the fluid, it will push fluid around to its surrounding, and it cause fluid to be moved and hence other bacteria will experience friction. This kind of interaction experienced by the bacteria is called hydrodynamic interaction, which is a long-range effect that decays with the inverse distance between the particles [16]. The computational method we will be using is called Brownian dynamics in which the particle dynamics is described by overdamped equation of motion with force and noise. The displacement is given as

$$r_i = r_i^0 + \sum_j \frac{D_{ij}^0 F_j^0}{kT}\Delta t + R_i(\Delta t), \tag{2.9}$$

where superscript ''0'' indicates that the variable is to evaluated at the beginning of the time step [17]. The displacement $R_i(\Delta t)$ is a random displacement with a multivariable normal distribution whose average value is zero, variance and covariance is $2D_{ij}^0\Delta t$. The diffusion matrix value D_{ij} can be given as

$$D_{ij} = \begin{cases} \dfrac{kT}{6\pi\eta a}\boldsymbol{I} : & i = j \\ \dfrac{kT}{6\pi\eta a}\left(\boldsymbol{I} + \dfrac{\bar{r}_{ij}\bar{r}_{ij}}{r_{ij}^2}\right); & i \neq j \end{cases} \tag{2.10}$$

which is also known as Oseen tensor [17].

Since we want to simulate bacteria swarming behaviour, we introduced driving force to the bacteria. The speed of bacteria movement is roughly between 20 µm and 30 µm [18, 19]. Because of this we test how much driving force each bacteria should experience in order to have the speed of 30 µm, the magnitude of the driving force is 0.15 pN.

Since we do not want the bacteria to overlap with one another, we introduce repulsive force term in the total force calculation. The repulsive force can be given as:

$$F_{\mathrm{repulsive}} = Ae^{B(r-R)} \tag{2.11}$$

where r is the scalar value of the displacement between two beads, $|\boldsymbol{r}_2 - \boldsymbol{r}_1|$, and R is the radius of the beads. We need to find the value of the constant A and B. Since we want the repulsive force to dominate in the short range and attractive force to dominate in the long range.

Next step, we increased the number of bacteria to more than three. We test different number of bacteria in different shape of starting position and in different temperatures. The effect of the limit on the repulsive force also investigated by changing the its value.

To measure the orientation of the bacteria and tell the alignment of the bacteria swarm, we use the order parameter, S, to measure the orientation of the bacteria as it was shown in Equation 2.12.

$$S = \langle P_2(\cos\theta)\rangle \tag{2.12}$$

where θ is the angle between the individual director and overall director [20], the angle brackets indicating an average over all the bacteria over multiple configuration drawn from the equilibrium distribution, and P_2 is the second order Legrendre polynomial defined by $P_2(\cos\theta) = \frac{1}{2}\left(3\cos^2\theta - 1\right)$. Therefore, for $S = 1$, all bacteria are pointing in the same direction.

The order tensor is given as:

$$Q_{\alpha\beta} = \frac{1}{2}\left\langle 3l_{\alpha,i}l_{\beta,i} - \delta_{\alpha\beta}\right\rangle_i, \qquad \alpha,\beta = x,y,z \tag{2.13}$$

where $l_{\alpha,i}$ and $l_{\beta,i}$, are components of the director of cell i expressed as a Cartesian unit vector, and $\delta_{\alpha\beta}$ is the Kronecker delta [20]. S is given by the largest eigenvalues of $Q_{\alpha\beta}$ with overall system director, which we don't actually need, given by the corresponding eigenvector.

Another parameter we will be measuring is the distance with the nearest neighbour, which can tells us how close are the bacteria moving together and tells us the whether there is collective motion. We use the predefined function in Python library ''scikit-learn'', and use the ball-tree-based neighbors searches algorithm to find the nearest neighbour of the bacteria.

3 Possible Applications of Collective Motion

Apart from the visual appeal from the collective motion, there is more possible future applications in understanding the principle behind it. One of which is the application in medicine, where using nanoparticles to deliver drugs.

Another potential application of studying collective motion is crowd control. In which case the organiser of a big event or ceremony want to prevent a riot take place.

A potential development in the applications of collective motion is on the study of social network.

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